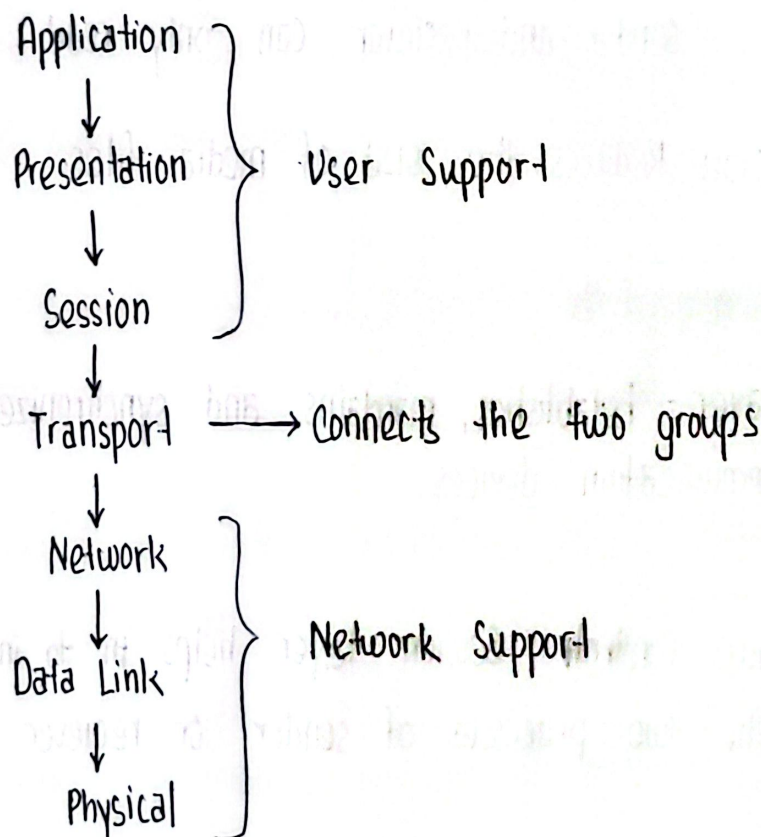


Data and Telecommunication

OSI Model : The International Organization for Standardization (ISO) developed the Open Systems Interconnection (OSI) reference model to organize communication functions into layers.

- * It is a reference framework, not one single protocol.
- * It helps interoperability between different networks



Application Layer - Enables user to access the network resources.

Service provided : ① File Transfer and Access Management (FTAM)

② Mail Services

③ Directory Services.

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Presentation Layer - It is connected with the syntax and semantics of the information exchanged between two systems.

Services Provided:

- ① Translation: Data is converted into a common format that is acceptable by both sender and receiver.
- ② Encryption: Converts data is converted into unreadable text so that sender and receiver can only read.
- ③ Compression: Reduces the size of media files.

Session Layer - Establishes, maintains and synchronizes the interaction among communication devices.

Services:

- ① Dialogue Control: Session layer helps in communication between the two processes of sender or receiver to take place.
- ② Synchronization: ~~If a send~~ The data is sent ~~in dif~~ divided into layers and then it is sent. It adds checkpoint to the data so that if an error occurs, the process can resume from that particular checkpoint. ~~instead~~

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Transport Layer - Responsible for process to process delivery of the entire message.

① **Port Addressing** : Sender sends data through a port number. When receiver receives it and again sends data back to the sender, the OS must determine the right process for it. This process is determined by source and destination port numbers.

② **Segmentation and Reassembly** : Divides the data in parts and is sent in that way. Then it is again reassembled.

③ **Connection Control** : Determines if it will be connection oriented or connection-less service.

④ **End-to-End Flow Control** : Determines suitable data transfer speed so that there is no data loss.

⑤ **Error Control** : Determines whether there is any error in the sent message.

Network Layer - Responsible for delivery of data from the original source to the destination network.

Services Provided -

① ^{Logical} ~~Local~~ addressing : Deals with IP addresses. Help router to determine the source and destination of a packet.

② Routing : Helps in finding out the best route ~~to~~ for the packet to be transmitted.

Data Link Layer : Responsible for moving data frames(data) from one node to another.

③ Framing : The DLL of the node groups the ~~be~~ bits of 0 and 1 into a frame.

④ Physical Addressing : Handle the MAC address.

⑤ Flow Control :

⑥ Error Control

⑦ IF Access Control : If two or more layers are connected to the same link, ~~DD~~ DLL helps to determine which device has control over the link at that time.

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Physical Layer - Responsible for transmitting bits over a medium (wired or wireless).

Services Provided -

① Physical characteristics of the media

② Representation of bits.

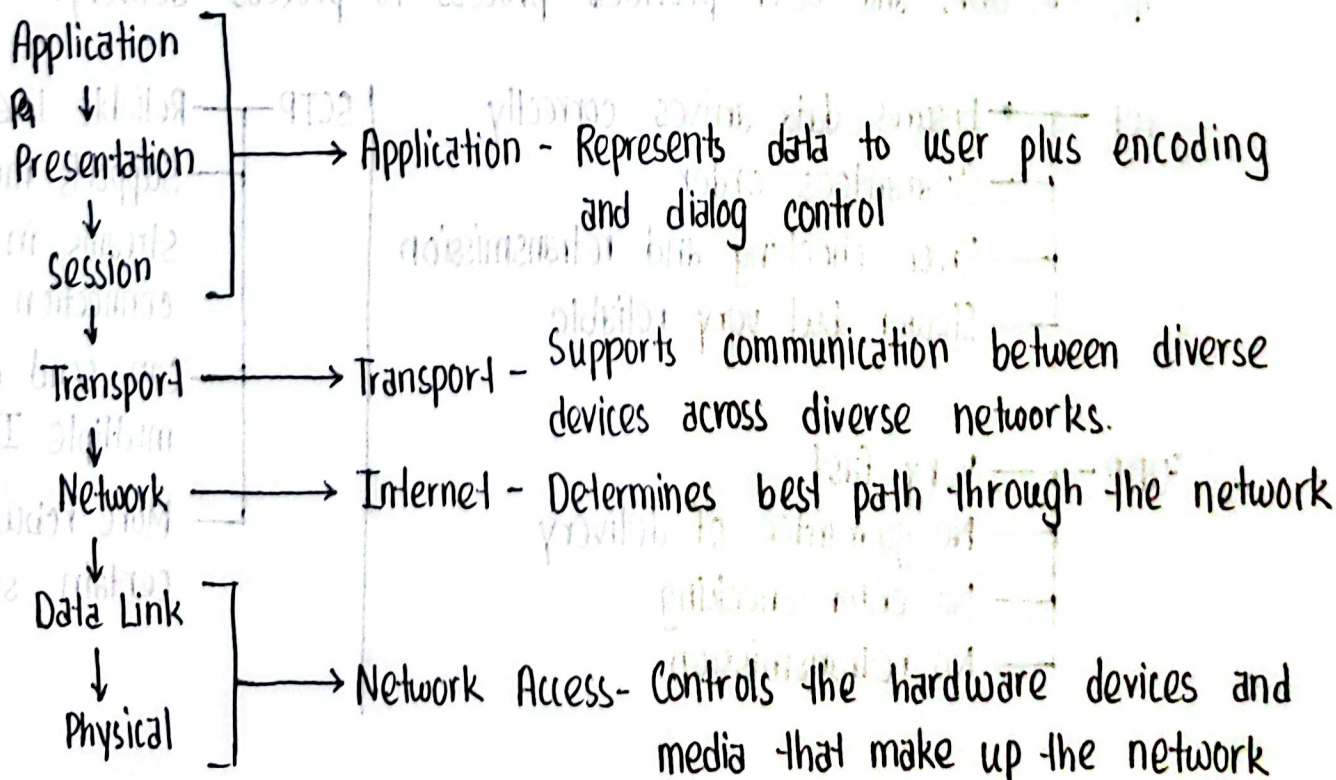
③ Data rate

④ Synchronization of bits

⑤ Line configuration

⑥ Physical Topology

OSI Model Vs TCP/IP Model-



TOPIC NAME:

Application Layer : HTTP • P (HyperText Transfer Protocol) - Protocol that powers data transfer on the web between browser and websites.

SMTP (Simple Mail Transfer Protocol) - Protocol used for sending emails

FTP (File Transfer Protocol) - Protocol for transferring files between computer and a server.

DNS (Domain Name System) - Translates website names into IP addresses so computer can find each other.

Transport Layer :

☐ IP provides host-to-host delivery

☐ TCP, UDP, and SCTP provides process-to-process delivery.

TCP

- Ensures data arrives correctly
- Guarantees order
- Error checking and retransmission
- Slower but very reliable

UDP

- Very fast
- No guarantee of delivery
- No order checking
- No retransmission

SCTP

- Reliable like TCP
- Supports multiple streams in one connection
- Can send data to multiple IPs
- More robust for certain systems

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Internet Network Layer :

IP — Assigns IP addresses to devices
— Routes data from source to destination

ARP — Finds the physical address (MAC address) of a device's IP

RARP — Asks for IP address of a MAC address (replaced by DHCP)

ICMP — Sends user reports

IGMP — Allows devices to join/leave multicast groups
— Used for streaming to multiple users

Network Access Layer :

